Smart Building Layout and Emergency Exit Planner

Semester-Wide Project

Course: Artificial Intelligence **Group 6**

Background and Motivation

Designing a building is not simply a matter of placing rooms inside an available area. A good layout must support the daily activities of its users, provide accessible movement, reduce unnecessary energy use, and allow people to leave safely during an emergency. These requirements may conflict with one another. For example, placing related rooms close together may improve convenience, but narrow corridors or poorly positioned doors may create congestion during evacuation.

Consider the design of a small hospital, office, school, or hostel. The building may require rooms of different sizes, accessible entrances, lifts, emergency exits, open spaces, service areas, and connections between related rooms. The number and purpose of some rooms may also change in the future. A meeting room may later become a laboratory, or an office area may need to accommodate more people. Therefore, the system should not generate only one visually attractive floor plan; it should produce a layout that remains functional, safe, and adaptable.

In this project, you have to develop a **Smart Building Layout and Emergency Exit Planner**. Given natural-language requirements, the system should generate possible building layouts, check spatial and safety constraints, evaluate evacuation performance, and explain the advantages and limitations of each proposal. The project should be implemented in a simulated environment and should not be treated as a replacement for professional architectural, structural, accessibility, or fire-safety approval.

The project is inspired by *Lang2Plan*, a multi-agent system for building-layout planning accepted in The Web Conference 2026 Demo Track (Pan et al. 2026). It is also related to *HouseTune*, which combines LLM reasoning with a specialised generation model to produce floor plans from natural-language descriptions (Zong et al. 2026). Recent research on reinforcement-learning-based evacuation planning further shows how building configurations and escape policies can be evaluated under changing fire and congestion conditions (Fang et al. 2026; Fitkau and Hartmann 2025).

Module 1: Search and Constraint Satisfaction

You have to represent a building layout using rooms, doors, corridors, stairs, lifts, exits, and their spatial relationships. Search algorithms such as Best-First Search, A*, Monte Carlo Tree Search, or another suitable method should be used to explore alternative room arrangements. You also have to formulate layout generation as a Constraint Satisfaction Problem. The constraints may include required room areas, room adjacency, separation of incompatible spaces, corridor width, accessibility, maximum travel distance to an exit, natural lighting, ventilation, and emergency-safety requirements. The system should distinguish between mandatory constraints and design preferences, detect infeasible requirements, and explain which constraints are responsible for the conflict.

Module 2: Construction and Evacuation Planning

You have to model both construction activities and emergency evacuation as planning problems. Construction actions may include preparing spaces, constructing walls, installing doors, placing utilities, and completing safety equipment, with suitable dependencies among these activities. Evacuation actions may include selecting an exit, moving occupants through corridors, avoiding dangerous areas, opening emergency routes, and guiding different occupant groups. The planner should construct valid action sequences and update them when an exit becomes blocked, congestion increases, a lift becomes unavailable, or the location of a hazard changes. Partial-order planning may be used because many independent construction or evacuation activities can occur simultaneously.

Module 3: Reasoning under Uncertainty

You have to model uncertainty related to room occupancy, crowd movement, fire location, smoke spread, equipment failure, and the availability of emergency exits. Bayesian reasoning or another suitable probabilistic method should be used to estimate the condition of different areas and update these estimates as new observations become available. For example, the system may receive uncertain information from smoke detectors, occupancy sensors, or simulated reports from building users. It should maintain alternative evacuation options when the safest route is uncertain and should avoid assuming that every exit, alarm, lift, or emergency device will remain operational.

Module 4: Reinforcement Learning

You have to formulate layout improvement or emergency-route selection as a Reinforcement Learning problem. For layout design, the agent may modify room positions, corridor widths, doors, stairs, exits, or the intended use of flexible spaces. For evacuation, the agent may direct occupants towards different exits, temporarily close a dangerous route, divide crowds across several paths, or update instructions as conditions change. The reward function may consider evacuation time, congestion, exposure to hazards, accessibility, construction cost, energy efficiency, usable area, and the number of layout constraints satisfied. You may implement Q-learning, Deep Q-Networks, PPO, multi-agent reinforcement learning, or another appropriate method and compare it with fixed layouts or static shortest-path evacuation strategies.

Module 5: Generative AI

You have to develop a GenAI-based designer agent that translates natural-language requirements into structured rooms, relationships, preferences, and constraints. For example, the user may request that classrooms remain close to laboratories, that all floors be wheelchair-accessible, or that office spaces can later be converted into meeting rooms. The system may contain specialised designer, accessibility, energy, evacuation, and constraint-checking agents. These agents should collaboratively generate and revise proposed layouts.

The LLM should help interpret requirements, ask clarification questions, suggest alternatives, and explain the final design. However, it should not independently decide that a layout satisfies legal or safety regulations. Such decisions must be verified by the CSP, planning, simulation, and rule-checking components. The final explanation should describe the important room relationships, accessibility features, emergency routes, expected evacuation performance, sustainability considerations, and any requirements that could not be fully satisfied.

Expected Outcome

The final system should demonstrate how search, constraint satisfaction, planning, probabilistic reasoning, reinforcement learning, and generative agents can be combined to create and evaluate building layouts. You should test the system using different building types and scenarios involving changing room requirements, high occupancy, blocked exits, fire in different locations, equipment failure, mobility-impaired occupants, and future changes in room usage.

A successful implementation should generate feasible layout alternatives, satisfy important functional and accessibility requirements, support safe evacuation, and clearly explain design trade-offs. The evaluation may consider constraint-satisfaction rate, usable floor area, average distance between related rooms, accessibility, estimated energy efficiency, evacuation time, congestion, hazard exposure, number of layout revisions, and adaptability to future changes.

References

Fang, Hongqiang et al. (2026). "Development of a tenability-based path planning model for building fire evacuations using reinforcement learning". In: *Journal of Building Engineering* 118, p. 115132.

Fitkau, Isabelle and Timo Hartmann (2025). "Applying multi-agent reinforcement learning for escape path planning in a fire evacuation environment". In.

Pan, Fangyong et al. (2026). "Lang2Plan: An LLM-Powered Multi-Agent System for Building Layout Planning". In: *Companion Proceedings of the ACM Web Conference 2026*, pp. 101–103.

Zong, Ziyang et al. (2026). "Housetune: Two-stage floorplan generation with llm assistance". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 40. 16, pp. 14059–14067.